

Venkata Sai Manoj Channa

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PROFESSIONAL SUMMARY

AI/ML Engineer with a strong foundation in full-stack development, skilled in building and deploying machine learning models and data-driven applications. Experienced with deep learning frameworks (TensorFlow, PyTorch), computer vision, and end-to-end pipelines spanning data preprocessing, model training, and production deployment.

EDUCATION

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering 2022 - 2026

Institute of Aeronautical Engineering, Hyderabad

CGPA: 6.6/10

TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript, React.js, Next.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs

Databases: MongoDB, MySQL, PostgreSQL

Programming Languages: JavaScript, TypeScript, Python

Tools & Platforms: Git, GitHub, Docker, Postman, VS Code

Core Concepts: Data Structures, Object-Oriented Programming, System Design Basics, API Design

PROJECTS

Speech Enhancement Using Spectrogram Denoising with Deep U-Net Architectures | [\[GitHub\]](#)

Python, TensorFlow, PyTorch, NumPy, Librosa

- Built a deep learning model for speech enhancement using spectrogram denoising.
- Implemented a U-Net architecture to remove noise and reconstruct clean audio signals.
- Published as a research paper (see Publications section).

Real-Time Helmet & Triple-Ride Detection | [\[GitHub\]](#)

Python, YOLOv3, CNN, TensorFlow, PyTorch, OpenCV

- Built a computer vision pipeline detecting helmet violations and triple-riding with 90%+ accuracy on test data; optimized to 25+ FPS for real-time traffic surveillance.
- Fine-tuned YOLOv3 on a custom-labeled dataset and implemented automated violation detection for law enforcement alert systems.

PUBLICATIONS

- "Speech Enhancement Using Spectrogram Denoising with Deep U-Net Architectures" – Published at ICISSC. | [Certificate](#)

EXTRA-CURRICULAR ACTIVITIES

- Served as an NSS Core Team Member, contributing to community service initiatives and college-level volunteer programs.
- Active participant in competitive programming contests and technical meetups/developer communities.